SYLVOPAST a multiple target role-playing game to assess negotiation processes in silvopastoral management planning

Abstract : After a brief description of the framework of the model developed to simulate vegetation dynamics, fire propagation and agents behaviour, the role-playing game rules are presented and related to the different points they are supposed to deal with : climatic hazard, animal grazing, forest management, grazing duty, financial support. The results of several sets of game sessions are analysed according to markers based on the main stakeholders viewpoints, to an evaluation of the negotiation process and to the way land was structured as a result of a step-by-step compromise between the players. Finally, general conclusions are drawn on how such type of role-playing games can provide a methodological framework to build up negotiation support tools and can be used with different categories of persons.

Key-words: Role-playing game, Negotiation, Silvopastoral management, Agent-based modelling, Multi-agent system

Introduction

Silvopastoral management planning is supposed to organise on a long-term perspective, the deliberate retention of trees with animals in interacting combinations for multiple products or benefits (Nair, 1993). Under Mediterranean conditions, forest operations are mostly developed for public benefit and involve a great number of motivations among which fire prevention, conservation and biodiversity, rural development and landscape management (Bland and Auclair, 1996). Generally exclusively built up by forest planners, forest management plans gave little account for silvopastoral management or failed because they were not based on a deep understanding of multiple use rules and constraints.

In order to help foresters to better integrate this dimension to their planning work a role-playing game (RPG) was developed as a didactical support to silvopastoral training programmes. Once tested with real partners (a farmer playing with the forester he has really to work with), the Sylvopast RPG rapidly became a means of enriching and improving the representation of the negotiations and interactions between livestock farmers and foresters involved in the management of the same forest. The idea was to place stakeholders in a prospective situation that confront them with new environmental and social conditions, following the approach developed by Mermet (1993) and Piveteau (1994). The originality here was to combine a modelling approach based on multi-agent systems with the RPG methodology tested by these authors (Barreteau et al, 2001; Bousquet et al, 2002).

But the original aspect of the approach was to build-up a tool flexible enough to permit to play with actively involved stakeholders such as the current users of the resource (local farmers and foresters), with potential regulators of the system (managers or administrators), with technical experts (extensionists, technicians) or with learners concerned with the topic (students, scientists).

Four fundamental questions were tackled:

- How can the evolution of a silvopastoral system governed by contrasting pastoral and forestry management rules be simulated in a ludic way ?

- How can a such a complex system be represented in such a simplified way that makes it easily shared by all the stakeholders ?
- How can the most viable type of co-operation between partners in the medium and long term be evaluated ?
- Does a unique optimal spatial structure satisfying the objectives of each stakeholder really exist ?

Methods

The Sylvopast RPG had to represent the spatial dynamics involved in silvopastoral development, and to stimulate the raising of original strategies for striking a compromise between livestock farmers and foresters. So it was developed by combining an ecological model with an agent-behaviour model interacting through a negotiation process.

The ecological model

The basic principle behind the design of the ecological model is that any Mediterranean forest can be defined by combinations of basic plant types (tree/shrub/grass). These combinations lead to eight vegetation structures whose attractiveness differs depending on the user (figure 1). Each vegetation type has its own natural dynamics running through the progressive steps of the natural succession. But this trend can be stopped by some catastrophic event such as a fire or run out by human activities such as grazing, chopping or sowing, or even accelerated by tree planting.

The pastoral value of the vegetation depends on vegetation structure and climate. To simplify, vegetation types without grass have no pastoral value; when grass is mixed with shrubs, the productivity decreases. Compared to pure swards (table 1), the association of tree and grass enhances the productivity when climatic conditions are dry because of better moisture conditions below the trees but depletes it a little when climate is rainy because of some lack of light (Etienne, 2000).

Table 1 : main markers on grazing potential according to vegetation structure

	Grass (G)	G + shrubs (S)	G + S + trees (T)	G + T
Dry period	15	5	10	30
Mild period	30	15	15	50
Rainy period	45	20	15	40

The agent-behaviour model

The silvopastoral management of Mediterranean forests implies the co-ordination between two main agents : livestock farmers and foresters. But it also frequently involves other important categories of agents such as hunters, hikers or firemen (Etienne, 1990). As a first step only the two first types were taken into account. The farmers were supposed to establish their farm near the forest and try to cover part of the requirements of their flock by grazing the areas they are allowed to by the foresters. The observation of real cases for many years lead to the fact that farmers choose their grazing unit according to 4 main criteria : the level of requirements of the flock, the available amount of forage, the climatic conditions and the travelling distance (Etienne, 1996; Etienne et al, 2002). They plan their production system at two scales of time : a weekly or monthly scale for the grazing calendar, a ten-year scale for the ewe life program. Concurrently, Mediterranean foresters plan their operations on a year basis for the labour calendar or fire prevention, and at a ten-year scale for tree regeneration and sylviculture. They schedule operations into some forest stands according to several goals. Regular clearings of the undergrowth are performed on specific zones strategically located to prevent fires from running across the forest or destroying the most productive stands. Selective thinning and clearing is applied to parts of the forest where biodiversity is looked after. Tree planting is developed in places where tree canopy is unsatisfactory or where a shift towards a more interesting dominant tree canopy is wished.

In order to transform these behaviour rules into game rules, each player was given the ability to do some specific or shared actions. The farmer-player moves his flock from one cell to another one, and can buy animals. The forester-player can modify the forest structure by using a set of three techniques : sowing (introducing grass) can be applied on any vegetation type without grass except rocks, clearing (getting rid of the shrubs) can be applied on any cell with shrubs and reforesting (planting trees) can be applied on any vegetation type without tree except rocks.

The game markers

To take their decisions or to argue during the negotiation, players need to share some markers and a common way of representation of the value of these markers. In Sylvopast RPG, markers were developed to define the vegetation structure, to evaluate the management effectiveness and to measure the cost-benefit efficiency.

First, the global spatial entity must be defined (Figure 1)). The initial landscape is a virtual square forest of 100 cells subject to a westerly wind, and with some probability of fire and climatic hazards. The three elementary colours represent the vegetation types dominated by only one plant type (blue for trees, red for shrubs and yellow for grass) and the other vegetation types are painted with the corresponding composed colour (e.g. orange for the overlapping of shrubs and grass). This marker is imposed by the model designer and must be rapidly learnt and adopted by the players at the beginning of the game. It permits to monitor changes in vegetation structure according to natural vegetation dynamics and technical operations. For instance, shrub encroachment occurs when a yellow cell changes to orange after 3 years without grazing; similarly, clearing applied on a brown cell changes its colour to green.

Management effectiveness is first linked with the way climatic variability and fire risk are taken under consideration all along the process of the game. For the climate, players know that a year can be dry, mild or rainy and how many dry, mild or rainy periods will occur during these types of years. But, as in the reality, they don’t know by advance which type of climate is going to occur : the type of year and the climate of a period are sorted by the computer. For the fire, the players can monitor the fire propagation index that is automatically calculated and summed up as the forest fire risk index or displayed on a map at the end of each turn. Obviously, as fire occurrence and setting place are difficult to foresee, they are randomly sorted by the model, fire being always set in a place encroached with shrubs (red, orange, brown or purple cells).

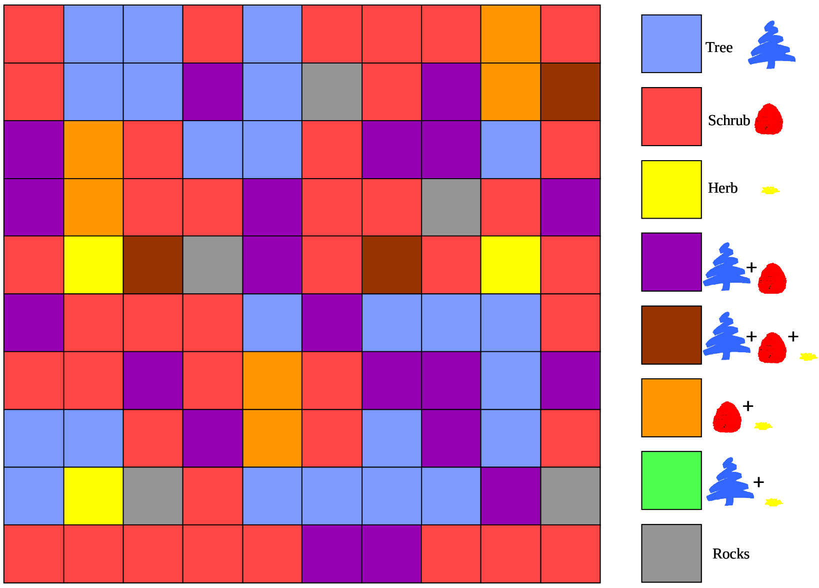

Figure 1 : The interface of the Sylvopast model and RPG. The colours correspond to combinations of basic plant types

Management effectiveness is also directly linked with the fulfilment of the objectives of each player. The farmer-player can monitor the grazing impact by checking the amount of food collected in the forest and the amount of energy wasted for travelling. He can also evaluate if the forest was improved by the technical operations by means of the pastoral value index that is automatically calculated as the sum of the average forage productivity of each cell. The forester-player can monitor two indexes that account for each of the goals he can aim at : the forest index gives an idea of the area covered by pure tree stands (number of blue cells), the landscape diversity index measures the global diversity of vegetation structures in the forest (note over 100).

Finally, all the technical operations have a cost which can be specific to a player (e.g. the grazing duty must be paid by the farmer) or shared (e.g. clearing of the undergrowth) and are planned every year. The players have to avoid too much expenses and to secure enough incomes so their capital will not reach unauthorised levels : the forester cannot run into debt and the farmer leaves definitely the forest if he is not able to cover by grazing the forest the maintenance requirements of 80% of his flock. The farmer incomes come exclusively from grazing while the forester ones are made of the grazing duty and of a government incentive. This incentive is determined by wildfires occurrence and the average state of improvement of the forest. Fire risk decrease, landscape diversity increase and forest area increase are considered as improvements. If the sum of the three markers is positive, the forester gets the incentive, if it is null or negative, he does not (table 2). After each wildfire, the forester gets a fixed financial support to help him to restore the forest.

Table 2: Mode of calculation of the average state of the forest

	Fire risk	Landscape diversity	Forest area	Average state
Turn 3	**200**	**22**	**21**	
Turn 4	**198**	**26**	**20**	
Turn 5	**198**	**28**	**19**	
Improvement 4/3	**+**	**+**	**-**	+
Improvement 5/4	**o**	**+**	**-**	**o**

The game rules

Shepherd player : A grazing duty corresponding to the flock size (1 per animal, starting at 250), has to be paid to the forester at the beginning of every round. During each round the shepherd moves his flock on k cells, k corresponding to the flock size divided by 25 (starting at 10) and sums up the feeding value of the cell to his capital minus the travelling cost. He selects the cells according to climate and vegetation as well as travelling distance. Any travelling (diagonal forbidden) of 3 to 8 squares costs 5, and of more than 8 squares costs 10. At the beginning of any new round, the shepherd can increase the size of his flock (each lot of 25 animals costs 25 and gives the right to graze a new square). But he will leave definitely the forest if his capital is lower than 80% of the flock size.

Forester player : The forester can choose among three goals : to reduce fire hazard, to increase the area of dense forest stands (blue cells) and to increase vegetation diversity. If he got more positive than negative trend of these 3 indexes, the global value of the forest is improved and he gets 250 from the Bank. In case of wildfire, a special grant of 500 is remitted to the forester in order to restore the burnt area.

Negotiation: At the end of each round, foresters and shepherds have 10 minutes to negotiate an improvement of the forest and to decide to support or not the corresponding operations. The way to negotiate is entirely free and any argument can be used by both partners to win the negotiation. In case of non agreement at the end of the negotiation, the forester gives the final cut.

The monitoring of the game

Recording: Previously designed as a classic game device, Sylvopast was run on a cardboard grid, with pins, adhesive coloured labels and false banknotes to be as ludic as possible. But this way of playing required a total dedication of the game master and impeded to record the major part of the actions. The shift to a computer device, permitted to visualise automatically the operations undergone during the game and to simulate their impact on vegetation dynamics. It also allowed to carry out automatic calculations and to keep a full record of all the strokes played and their sequence. Meanwhile, the game master and his assistants (if several games are played simultaneously) were free to observe the negotiation process and pick up the essential of the players behaviour.

Debriefing: The results are collectively analysed immediately at the end of the game. This debriefing is divided into four steps: a comparison of the evolution of the main markers, a summary made by the players of the main points of the game they just end up, a contradictory comment by the game master based upon an accelerated replay of the game, and a comparison with other games selected as typical examples of space organisation, behaviour or negotiation processes, the players can benefit from.

Results

Markers: Five markers were used to monitor the livestock farmer ability to push the negotiation towards his own interest. Two were dealing with the transformation of the forest towards high productive grazing lands (pastures in yellow or wooded pastures in green), one was a global evaluation of the grazing value of the forest and the two last one referred to the capitalisation process through the flock size or the cash flow.

Tables 3 and 4 presented some results from 4 contrasting types of agreement between both partners of the negotiation. Cases 1 & 2 raised out from a balanced negotiation but in one case, priority was given to cash flow while, in the other one, it was given to flock size. Case 3 corresponded to negotiations dominated by the shepherd while case 4 illustrated negotiations dominated by the forester. If we consider that the current number of games already played is statistically significant, and that the farmers and foresters who participated to these games are representative of the stakeholders concerned by the topic, case 4 was corresponded to 30 % of the games, while cases 1, 2 and 3 occurred in respectively 25%, 25% and 20% of the games.

Table 3: Value after ten rounds of five markers on the livestock farmer strategy compared with their state at the beginning of the game

	Yellow cells	**Green cells**	**Grazing value**	**Flock size**	**Capital**
Initial value	3	0	15	250	500
Farmer 1	6	7	35	325	680
Farmer 2	6	6	35	400	490
Farmer 3	8	10	47	450	1350
Farmer 4	4	5	23	250	420

Five other markers were used to monitor the forester ability to push the negotiation towards his own interest. Two were dealing with the forest sensitivity to fire (the burnt area and the global fire hazard), one concerned the transformation of the forest towards high productive stands (blue cells), one was an ecological assessment of the forest development through an index of vegetation diversity and the last one gave an idea of the level of investment into the forest management according to cash flow.

Table 4: Value after ten rounds of five markers on the forester strategy compared with their state at the beginning of the game

	Burnt cells	**Blue cells**	**Fire hazard**	**Forest diversity**	**Capital**
Initial value	0	23	216	12	500
Forester 1	23	10	179	12	325
Forester 2	12	20	172	30	450
Forester 3	1	25	160	28	250
Forester 4	14	29	167	16	275

Land-use patterns: Several games lead to a final result considered as satisfactory according to both the productive markers and the vegetation map. The pattern of the spatial distribution of vegetation types were quite different among this set of satisfactory solutions. When the negotiations were dominated by the shepherd, an important increase in grasslands (yellow cells) and wooded grasslands (green cells) was observed (case 3 on Table 3). But this domination lead to two different land-use patterns (Figure 2). Some farmers developed grazing units composed of 4 to 6 green and yellow cells. In such a configuration, they first grazed all the cells available in one block moving from green to yellow cells according to climatic conditions and then moved their flock from one block to one of the nearest ungrazed one. Other farmers developed a chain of alternating green and yellow cells. In this case, they organised a grazing circuit moving along the chain, the colour of the cell to be grazed being chosen according to climatic conditions.

When the negotiations were dominated by the forester, an important increase in woodlands (blue cells) and other wooded areas (green, brown or purple cells) was observed (case 3 and 4 on Table 4). But this domination lead to different land-use patterns according to the main goal assigned by the player to the forest management plan (Figure 3). Some foresters gave priority to timber production and limited grazing to strategic cells in relation to fire propagation. In such a configuration, they developed pure forest stands of around 10 cells and allocate to the shepherd the minimum of cells with grass (green, yellow and orange cells) necessary for him to survive. Other foresters focused their management on biodiversity enhancement. In this case, they operated on cells occupied by common vegetation types in order to develop original vegetation structures (green, orange and brow cells). But to maintain these new structures, they had to put pressure on the shepherds and oblige them to graze low pastoral value cells.

Fire prevention strategy had also a strong impact on land-use pattern (Figure 4). When linear fuel-breaks were preferred, vegetation types with grass and grazing were limited to a vertical “green and yellow” belt that separated the land in two major compartments. When alveolar gaps were selected, openings were located at the eastern edge of the woodlands to be protected and were managed in order to permit a permanent grazing. The first strategy was considered as totally effective or non-effective according to the place where the fire was set, but was chosen by players who look for a management plan quite simple to implement. The second strategy was always considered as partly effective because it permits to avoid any big wildfire, but it required more skills as it depends on the success of the negotiation with the shepherd and asked for a more elaborated argumentation.

All these examples demonstrated clearly that there is not one optimal solution to the management of this virtual Mediterranean forest, but a set of solutions corresponding to desired future conditions. And the many Sylvopast sessions permitted to represent these desirable future conditions as the conjunction of a range of values for observable indicators (Nute et al, 2000).

Discussion

Simplification of the model: The change in the rules regarding vegetation dynamics primarily concerned long-term processes such as the colonisation of abandoned areas by trees or the substitution of forest species, which were not included in the RPG. This permitted to run the game faster but limited to 10 the number of rounds of one game. Opposite to the model, the negotiating methods were left entirely open, with the only constraint being the need to develop a way of recording the sequence of actions on the part of each agent. The method consisting in asking each player to justify each of his actions in writing was abandoned, since it excessively reduced spontaneity. The new method chosen involves a confrontation between two pairs of players (i.e. two players for the same role) in order to oblige the players to explain their choices and discuss them with their partner, under the watchful eye of an observer. The observer also remains with a record of any action undergone during the game, which facilitates monitoring, carries out automatic calculations and keeps a full record of all the strokes played and their sequence.

A multiple target role-playing game: The RPG was developed with a view to four objectives resulting from four different targets:

- In playing with managers, the aim was to create a teaching aid to enable forestry practitioners to appreciate the constraints on herdsmen, and vice versa. The game thus enabled players to put themselves in the other players' shoes and better appreciate their needs and difficulties.
- In playing with stakeholders, the aim was to "switch" the development operation based on a broad sample to an area they were used to managing or evaluating. In this case, the game served to identify the main spatial organisation strategies developed and to establish a typology of the negotiating tactics used.
- In playing with students in agronomy, forestry or animal production science, the game was used to evaluate the level of understanding of the interactions into sylvopastoral systems, and to measure their ability to integrate several scales of space and time. With students from other disciplines such as management planning or environmental development, the game served to illustrate and analyse the negotiation process in a multiple use scope and to point out how difficult mediation can be in such a context.
- In playing with amateurs with no knowledge of either animal production or forestry, the aim was to test the clarity of the rules of the game, their ease of application and their ability to express clearly the problem of silvopastoral development. The game in this case served to make players aware of the complexity of forest development, and was a means of encouraging everyone to test, and even improve, his own natural resource management capacities.

The strong interest and even enthusiasm shown by the different categories of users of the game in our many tests (more than 80 games already played) showed that at least some of the objectives had been achieved. When real managers play the game, it was interesting to reverse the roles to better understand the requirements and difficulties of the other. When professionals play the game, it was easy to check the pertinence of the representations, to validate the technical rules and to elaborate a typology of negotiation strategies. The RPG demonstrated also to be generic enough to be used with students or with totally ingenuous people who, by playing a brand new role, learnt to test and even to develop their skills of managing natural resources under strong environmental constraints.

Limits of the method: However, several questions remain. In terms of implementation, the use of a computer, made compulsory by the need to record automatically as much information as possible, may be considered as an obstacle by some players compared with other classic game devices such as playing cards, coins, pawns, dice, etc. Previous tests done with this type of tools that are seen as more funny, showed that they stimulate the desire to win rather than the desire to act appropriately, and permitted to cheat more easily than in the computerised version.

As regards the categories of players, the first games were exclusively played by participants with the same level of knowledge in order to avoid a bias related to strong social or cultural differences between players. But after some tests, it was also considered interesting to mix the categories in order to increase the range of experience and rationales.

One of the main interesting points in the results coming out from the available set of games was that there is no one optimal solution to the natural resource management problem. Several games leading to contrasting spatial configurations were considered as satisfactory by both players and fitted perfectly with their objectives. This encourages to test a series of games where the players search for an optimum solution and replay a new game meanwhile the final result is considered unsatisfactory by one of the two players. In the first round, the RPG gives an idea of the current state of knowledge and power of the partners. In the following rounds, it points out their capacity of adaptation and flexibility to find a compromise towards a shared management objective. Presently this solution is only tested with a single person playing both roles at once again and again until he finds the most satisfactory solution. Anyway, the optimal configuration of the RPG is 2 shepherd-players facing 2 forester-players in order to oblige the players to orally express their tactics and for the observer to write them down.

Conclusion

RPG can be considered as an alternative to improve stakeholders involvement in participatory natural resource management, mainly when the scale of action mismatch reduces the chance of achieving the participation goals (Hare at al, 2002). Recent experiments on agent-based participatory modelling applied to water or land use management demonstrated that there are different ways of coupling MAS and RPG (D'Aquino et al, 2002). In some cases, the model simulates the dynamics of a resource and provides a dynamic basis for a negotiating game, in others the scientists' model has been converted into a role game with a view to validating or more clearly explaining stakeholder strategies, while in the last RPG is used for the collective construction of a MAS, which is implemented and supports the discussion on management scenarios.

Sylvopast RPG was developed under the second framework. The feed-back with the MAS was crucial because the RPG was conceived as a blueprint of the model but aimed at making easy to put the negotiation strategies into words and to understand the operations and decisions made by the agents. But as it demonstrated its ability to be easily and effectively used by learners it also fitted into the first framework as an entertaining negotiating game. It would be interesting to check if it could be used without any change as a support to a collective construction of a fire prevention management model.

The RPG approach permitted to identify and many times resolve conflicts between goals, and during the debriefing stage, to test current or projected situations for goal satisfaction. The proposed process is firmly grounded in the body of multi-objective decision analysis and RPGs demonstrated to be an interesting tool to structure goal hierarchy, define desired future conditions, select interesting management alternatives, and build up scenarios accounting for deep social interactions (Rauscher et al, 2000).

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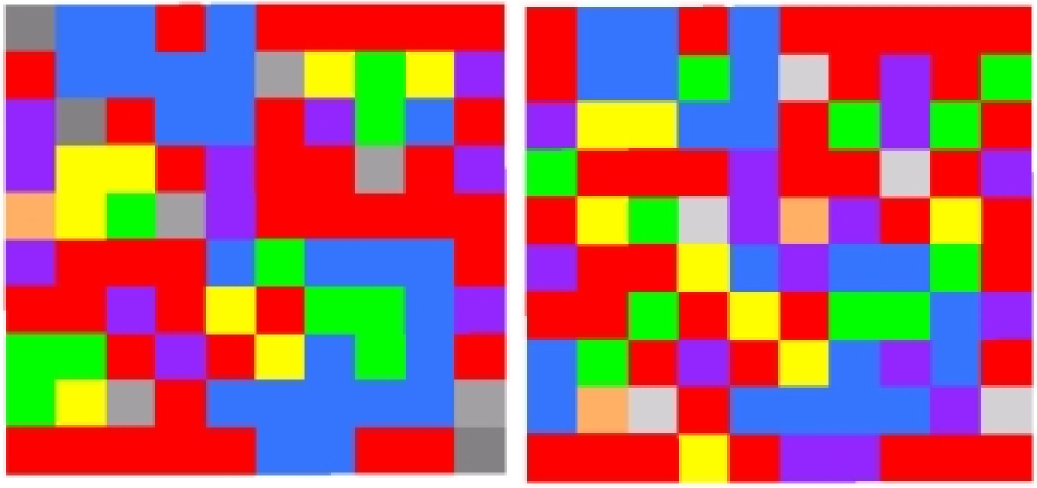

Figure 2 : Land-use patterns arising from negotiations dominated by sheperds, grazing unit (left) versus grazing circuit (right) strategy (colour legend on figure 1)

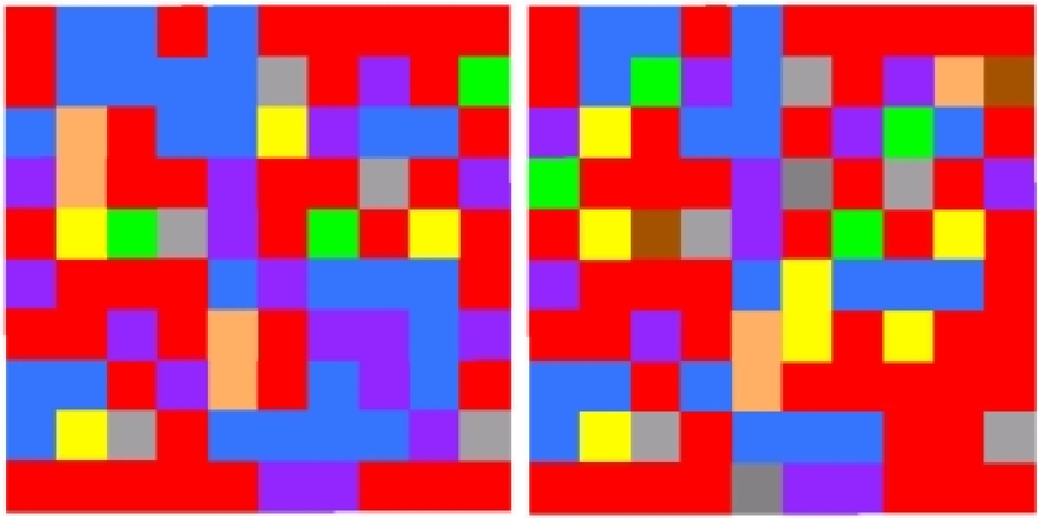

Figure 3 : Land-use patterns arising from negotiations dominated by foresters, for timber production (left) versus biodiversity enhancement (right) strategy (colour legend on figure 1)

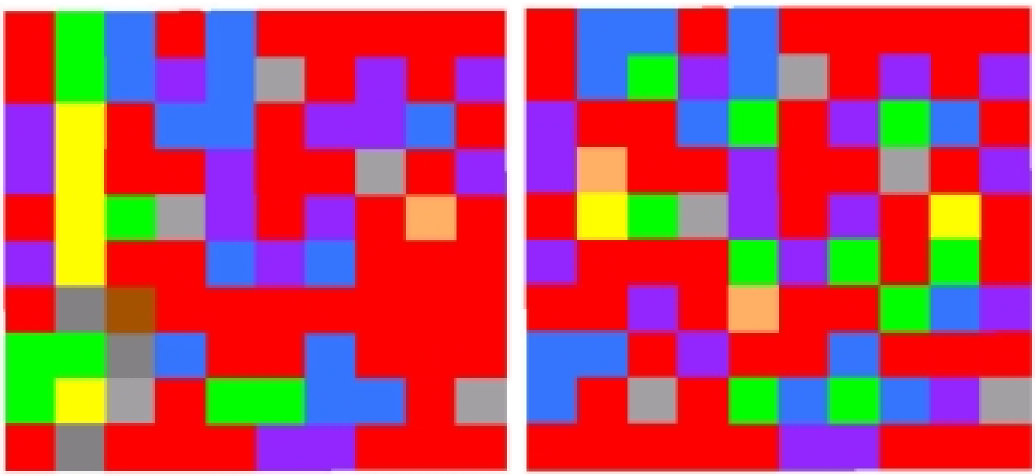

Figure 4 : Land-use patterns according to 2 contrasting fire prevention strategies, linear fuel-break (left) versus alveolar gaps (right) (colour legend on figure 1)